BT3041 – Analysis and Interpretation of Biological Data

Assignment 2

Instructions

Please provide your analysis results in a report format, specifically answering each question with relevant plots, etc. Also, state any assumptions made clearly in the report. Attach the Google Drive link to your software code (Python) used to perform the calculations for the report.

Submit your completed assignment report and supplementary files (if any) in Turnitin (Link for submission will be shared separately) by 10/04/2026 23:59:00. Any late submissions beyond this timeline will incur a penalty of 10 marks per hour. Obviously, if you submit it later than 10 hours, the entire assignment is considered forfeited.

Section 1: Regression **Marks: 40**

You are provided with a maize cultivation dataset comprising 1,658 observations across 14 predictor variables. These encompass environmental parameters, topographic features, soil physicochemical and fertility attributes, and farm management practices. Develop a regression model to predict maize yield (target variable) as directed below, enabling data-driven insights for optimized agricultural planning and yield forecasting.

Dataset

a) Split the dataset into Training, validation, and test datasets in a 60:20:20 ratio. [3]
b) Build linear regression models using all possible subsets of the 14 features from the training dataset. For each model, compute the Residual Sum of Squares (RSS) and plot RSS against the number of features included in the model. Identify and report the top three best-performing models based on AIC. [8]
c) For your top 3 models from (b), perform 5-fold cross-validation on the training dataset to assess its stability. [3]
d) Evaluate your top 3 models' performance on the validation dataset using metrics such as RMSE, MAE, and R^2. [3]
e) Using the training data, build L1- and L2-regularisation-based regression models. [4]

f) Perform 5-fold cross-validation on the training dataset to fine-tune the model's regularisation parameter developed in (e). [4]

g) Calculate the regularised model's performance on the validation dataset using RMSE, MAE, and R^2. [3]

h) Evaluate the performance of models developed in (d) and (g) in the test dataset using different metrics: RMSE, MAE, and R^2. [4]

i) Based on the metrics from parts (d) and (g), compare your regression models and comment on their relative accuracy. Additionally, based on your analysis, identify and suggest the key parameters farmers should carefully manage to optimise maize cultivation and maximise yields. [8]

Section 2: Classification **Marks: 60**

You wish to further utilize the maize cultivation dataset to categorize yield status into low, moderate, or high classes, using the environmental parameters, topographic attributes, soil physicochemical and fertility properties, and farm management practices as predictor variables. Categorical features such as soil type and irrigation method are also available. Develop multiple class classification models as instructed below to predict maize yield status from these features.

Dataset

Note: As Scikit-learn does not support categorical variables for classification, convert the categorical variables into numerical as suggested in the table below.

Numerical Notation	0	1	2	3
Soil Type	Clay	Sandy	Silt	Loamy
Irrigation Type	Canal	Rainfed	Sprinkler	Drip
Yield Status	Low	Moderate	High	-

a) Split the dataset into Training, validation, and test datasets in a 60:20:20 ratio. [8]

b) Build a decision tree classifier using the training data using shannon entropy criterion. [6]

c) Perform 5-fold cross-validation for the decision tree classifier using the training set. [6]

d) Evaluate the decision tree classifier using the validation dataset with: Confusion matrix, Precision, Recall, and F1-score (micro and macro averaged), ROC curve & AUC. [8]

e) Similarly, build a k-Nearest Neighbours classifier using the training data. Select the optimal k by plotting cross-validation accuracy versus k. [8]

f) Perform 5-fold cross-validation for the kNN classifier using the training set. [6]

g) Evaluate the kNN classifier model using the validation set with: Confusion matrix, Precision, Recall, and F1-score (micro and macro averaged), ROC curve & AUC. [8]

h) Report the accuracy of the decision tree classifier and the kNN classifier using the test dataset. Based on your observations, propose your best classifier. [10]